



Python

Python



- Guido Van Rossum is the creator of Python
- Released in 1991
- Python is an easy programming language, using python the complex programs can be written
- Python is a popular programming language





Why should we learn
Python? 

DATA SCIENCE & AI
SIVA RAO KRISHNA



Advantages of Python

Open Source : We can use python software without any license and its freeware.

Simple Syntax

- Readability
- Python syntax is similar to the English language
- Python syntax doesn't relies on curly-brackets & semi colon
- we no need assign datatype while assigning variable (**Dynamic Type**)

Interpreted

- Python is an Interpreter language, where Programs can be executed directly without a compiler.
- It will execute code directly, without compiling a program into machine-language instructions.



Differences between an interpreter and a compiler

Interpreter	Compiler
Executes program by taking one statement at a time	Translates the entire program at once into machine code
Error is reported as soon as the first error is encountered . Won't show the next set of errors if the existing one isn't solved	Errors are reported after the entire program is checked
Easier to debug	Difficult
No intermediate object code is generated, hence memory efficient	Generates intermediate object code which further requires linking, hence requires more memory
Overall execution is slower	Faster



Advantages of Python

High-level

- Programmer friendly, we not required to concentrate low level activities like memory management and security etc
- Python can be treated in a procedural way, an object-oriented way or a functional way.

Portability

- designed to be more or less independent of a particular type of computer, human-readable friendly
- Python works on different platforms (Windows, Mac, Linux, Raspberry Pi, etc)

Robust Standard Library

It contains no. of libraries, allows developers to write programs with fewer lines than some other programming languages.

Python



Core Python: Common for everyone who works on python

+

Python as per Roles in various fields

DATA SCIENCE & AI
SIVA RAMA KRISHNA



Applications of Python

- **Machine Learning** (using Sklearn, NumPy, Pandas, SciPy)
- **Artificial Intelligence**
 - Deep Learning (libraries like TensorFlow, Keras, Pytorch)
 - Natural Language Processing (Nltk, Spacy)
 - Image & Video Processing (using OpenCV)
 - Audio Processing (using PyAudio)
- **Web Applications**
 - Web frameworks such as Django
 - Micro-Frameworks such as Flask.
- **Games** (PySoy -> 3D game engine,
PyGame -> library for game development)
- **Web Scraping** (Requests, BeautifulSoup)



Python used in various fields

Python can be used in different ways such as:

- Software developers
- Data Analysts
- Data scientists
- Hackers
- Game Designers
- Scientists ,etc.

Python



HIGH-LEVEL

SCRIPTING

GENERAL-PURPOSE





Various Methods to Execute Python

DATA SCIENCE & AI
SIVA RAMA KRISHNA



Python can be executed in various ways

1. Python (command prompt)
2. Python IDLE
3. Run file in Python IDLE

with IDE (integrated Development Environment)

1. Jupyter Notebook
2. Spyder
3. Pycharm
4. Visual Code

WhatsApp channel:

<https://whatsapp.com/channel/0029Var9Wub>

Module1: Concepts

Topic1: Arrays

- 1. Definition & Declaration**
- 2. Array Initialization**
- 3. Accessing Elements**
- 4. Updating Elements**
- 5. Array Length / Size**
- 6. Traversing an Array**

24. $O(n \log n)$ time complexity

25. $O(1)$ space complexity

26. $O(n)$ space complexity

27. $O(n^2)$ space complexity

Topic2: Strings

28. String declaration and initialization

29. Immutability

30. concatenation

31. `length()` method

32. `charAt()`

33. Read the characters from left to right

34. Read the characters from right to left

35. `toCharArray()`

36. converting char array to String

37. `indexOf()`, `lastIndexOf()`

- 51. Min Heap development with
- 52. Max Heap development with
- 53. Lambda expressions.
- 54. Comparable
- 55. Comparator
- 56. List features and different
- 57. CRUD operations in the List
- 58. Set features and different
- 59. CRUD operations in the Set
- 60. Map features and different
- 61. CRUD operations in the Map
- 62. CRUD operations in the Priority Queue
- 63. CRUD operations in the Deque

Topic4: Recursion

- 64. What is recursion
- 65. Direct recursion
- 66. Indirect recursion
- 67. Tail Recursion

